

PRD

CarSwap Platform - Product Requirements Document (PRD)

Key sections include:

- Executive summary and vision
- Detailed POV explaining the opportunity and why now
- User personas (Enthusiast Eric and Dealer Dave)
- Complete MVP feature requirements with acceptance criteria
- Technical architecture recommendations
- Go-to-market strategy with timeline
- Risk analysis with mitigations
- 12-month roadmap with milestones
- Financial projections

The document is designed to:

- Guide your development as a solo builder
- Help communicate vision to potential co-founders
- Serve as a reference for prioritization decisions
- Track progress against milestones

A few highlights:

- MVP is scoped for 6-8 weeks of part-time development
- Emphasizes validation before building (landing page, manual matching)
- Clear success/kill criteria at each phase
- Realistic about risks (fraud, chicken-and-egg, legal)

Document Information

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Table of Contents

1. Executive Summary
 2. Point of View (POV)
 3. Problem Statement
 4. Goals & Success Metrics
 5. User Personas
 6. Product Overview
 7. Feature Requirements (MVP)
 8. Feature Requirements (Future)
 9. Technical Architecture
 10. Go-to-Market Strategy
 11. Risks & Mitigations
 12. Timeline & Milestones
-

1. Executive Summary

Product Name: CarSwap (working title)

Vision: Create the first dedicated social platform for car enthusiasts to discover, post, and facilitate direct vehicle exchanges - streamlining what is currently a fragmented, inefficient process.

Mission: Connect car lovers who want something new with others looking for what they have, reducing transaction friction and creating a community around the joy of vehicle ownership.

Target Launch: Q2 2026 (Beta), Q3 2026 (Public)

Key Differentiator: Unlike traditional classifieds or trade-in services, CarSwap is a visual, social-first platform designed specifically for direct peer-to-peer vehicle exchanges with cash differentials.

2. Point of View (POV)

The Opportunity

Market Observation: Car enthusiasts typically change vehicles every 2-3 years, driven by desire for new experiences rather than necessity. The current process is painful:

- List car on multiple platforms (Facebook Marketplace, Craigslist, AutoTrader)

- Field tire-kickers and lowball offers
- Negotiate sale
- Search for new vehicle
- Negotiate purchase
- Two separate title transfers, insurance changes, test drives

Customer Insight: "I'd love to trade my Elantra N for a C63 AMG, but finding someone who wants exactly what I have feels impossible. I end up going the traditional route even though it's slower and more expensive."

The Gap: No dedicated platform exists that:

- Makes vehicle swap discovery visual and social
- Focuses specifically on enthusiast-to-enthusiast exchanges
- Allows for flexible matching (direct swap + cash differential)
- Creates a community around the swap lifestyle

Why Now?

1. **Social commerce maturity:** Users comfortable with peer-to-peer transactions (Poshmark, StockX, FB Marketplace)
2. **Visual discovery:** Instagram/TikTok have trained users to browse visually for inspiration
3. **Enthusiast market growth:** Car culture is thriving (YouTube channels, Cars & Coffee, modification culture)
4. **Economic conditions:** Used car values remain high, making swaps economically attractive
5. **Technology accessibility:** Modern frameworks enable rapid MVP development

Why This Will Work

Network Effects: Each listing creates value for browsers; each browser creates potential demand for posters

Low barrier to entry: \$3-5 listing fee is impulse-purchase territory

Emotional appeal: Swapping feels exciting and community-driven vs. transactional selling

Niche focus: Serving enthusiasts specifically vs. trying to be everything to everyone

3. Problem Statement

User Problem

Car enthusiasts who want to change vehicles face a fragmented, time-consuming, and expensive process involving multiple platforms, negotiations, and transactions.

Current Alternatives & Their Limitations

Alternative	Limitation
Traditional sale + purchase	Two separate transactions, high friction, dealer markup on purchase
Dealer trade-in	Significant value loss (20-30% below market)
Facebook Marketplace/Craigslist	Not specialized, poor discovery, safety concerns
Enthusiast forums	Manual, unstructured, limited reach
Turo/sharing platforms	Rental model, not ownership exchange

Why Existing Solutions Fail

- **Not visual/social:** Text-heavy listings lack inspiration
 - **Not specialized:** Generic classifieds don't understand swap dynamics
 - **No community:** Transactional only, no identity or belonging
 - **Poor matching:** Manual searching, no algorithmic suggestions
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4. Goals & Success Metrics

North Star Metric

Monthly Active Swaps (MAS): Number of successful vehicle exchanges facilitated per month

Primary Metrics (MVP Phase)

Acquisition:

- 1,000 registered users by end of Month 3
- 500 active listings by end of Month 4

Engagement:

- 30% of users return within 7 days
- Average session duration: 5+ minutes
- 50+ messages sent per day

Conversion:

- 10% of visitors create an account
- 50% of account holders post a listing
- 5 verified swaps in first 3 months

Secondary Metrics

Quality Indicators:

- Average listing has 5+ photos
- 80%+ of listings include detailed description
- User-reported satisfaction score: 4+/5

Community Health:

- 20% of users save/favorite listings
 - User-generated content (swap stories, photos)
 - Repeat usage (users who complete swap return to post again)
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5. User Personas

Primary Persona: "Enthusiast Eric"

Demographics:

- Age: 25-40
- Income: \$60K-\$120K
- Location: Urban/suburban US

Psychographics:

- Passionate about cars (follows car YouTube, attends Cars & Coffee)
- Views car as expression of identity, not just transportation
- Active on social media (Instagram, Reddit, TikTok)
- Tech-savvy, comfortable with online transactions

Pain Points:

- Gets bored with current car after 2-3 years
- Traditional selling process is tedious
- Dealer trade-ins offer terrible value
- Finding the "right" next car takes months

Goals:

- Discover interesting swap opportunities
- Connect with like-minded enthusiasts
- Minimize transaction costs and time
- Experience variety in car ownership

Quote: "I love my car, but I'm ready for something new. I just wish I could trade it like Pokemon cards."

Secondary Persona: "Dealer Dave"

Demographics:

- Age: 35-55
- Role: Small independent dealer or car flipper
- Volume: 3-10 cars per month

Psychographics:

- Entrepreneurial, always looking for inventory
- Understands value of alternative sales channels
- Willing to pay for lead generation

Pain Points:

- Auction prices are competitive
- Needs consistent inventory flow
- Traditional advertising is expensive
- Wants access to motivated buyers

Goals:

- List multiple vehicles efficiently
- Reach qualified buyers
- Turn inventory quickly
- Build reputation

Quote: "If I can reach serious car people without spending thousands on ads, I'm in."

6. Product Overview

Core Value Proposition

For Car Enthusiasts: "Discover your next car by trading what you have - the visual, social platform for vehicle swaps"

For Dealers: "Reach qualified car enthusiasts actively looking to upgrade"

Product Principles

1. **Visual First:** Every interaction should be image-driven and inspiring
2. **Community Over Transactions:** Build belonging, not just a marketplace
3. **Transparency:** Clear badges for dealers vs. private, honest about risks
4. **Simplicity:** Remove friction, handle complexity behind the scenes
5. **Trust Through Openness:** Users handle their own risk; we facilitate discovery

User Journey (Happy Path)

1. **Discovery:** User browses visual feed of swap listings
 2. **Inspiration:** Sees a car they want, gets excited about possibility
 3. **Connection:** Messages poster to discuss swap details
 4. **Negotiation:** Users discuss vehicles, cash differential, logistics (off-platform)
 5. **Exchange:** Users meet, inspect vehicles, complete swap (off-platform)
 6. **Celebration:** User shares swap story, posts new listing for next swap
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7. Feature Requirements (MVP)

Phase 1: Core Platform (Weeks 1-6)

1.1 User Authentication

- **Priority:** P0 (Must Have)
- **Description:** Users can create accounts and log in
- **Requirements:**
 - Email/password registration
 - Google OAuth sign-in
 - Email verification
 - Password reset flow
- **Acceptance Criteria:**
 - User can create account in under 60 seconds
 - Session persists across browser sessions
 - Secure authentication (hashed passwords, JWT tokens)

1.2 Create Swap Listing

- **Priority:** P0 (Must Have)
- **Description:** Users can post what they have and what they want
- **Requirements:**
 - Upload 3-10 photos (min 3 required)
 - Enter vehicle details:
 - What I have: Year, Make, Model, Trim, Mileage, Location (city/state)
 - What I want: Make, Model preferences (open text)
 - Cash differential range (e.g., willing to add \$5K-\$10K)
 - Description field (500 char max for MVP)
 - Location-based (zip code for general area)
- **Acceptance Criteria:**
 - Listing creation takes under 3 minutes
 - Photos are compressed/optimized automatically
 - Location is validated against US zip codes
 - Listing goes live immediately after payment

1.3 Browse Feed

- **Priority:** P0 (Must Have)
- **Description:** Instagram-style visual feed of swap listings
- **Requirements:**
 - Infinite scroll of listings
 - Large, prominent photos
 - Key info overlay: Make/Model, Location, "Want: [X]"
 - Default sort: Most recent first
 - Works on mobile and desktop
- **Acceptance Criteria:**
 - Feed loads in under 2 seconds
 - Images are high quality but optimized
 - Smooth scrolling on mobile
 - No listings older than 90 days shown

1.4 Basic Search & Filters

- **Priority:** P0 (Must Have)
- **Description:** Users can find relevant listings
- **Requirements:**
 - Search by make/model
 - Filter by:
 - Location (within X miles of zip code)
 - Vehicle type (sedan, SUV, sports car, truck)
 - Price range (cash differential)
 - Clear all filters option
- **Acceptance Criteria:**
 - Search returns results in under 1 second

- Filters can be combined
- Results update without page refresh

1.5 Listing Detail View

- **Priority:** P0 (Must Have)
- **Description:** Full information about a specific listing
- **Requirements:**
 - Photo gallery (swipeable on mobile)
 - Complete vehicle details
 - Seller profile preview
 - "Message Seller" button
 - "Save" button
 - Posted date
 - View count
- **Acceptance Criteria:**
 - All images load progressively
 - Contact button is always visible
 - Works seamlessly on mobile

1.6 Direct Messaging

- **Priority:** P0 (Must Have)
- **Description:** Users can communicate about potential swaps
- **Requirements:**
 - Real-time chat interface
 - Message threads per listing
 - Push notifications for new messages (web/mobile)
 - Photo sharing in messages
 - Report/block functionality
- **Acceptance Criteria:**
 - Messages deliver within 2 seconds
 - Users can see when other person is typing
 - Message history is preserved
 - Blocked users cannot contact you

1.7 User Profile

- **Priority:** P0 (Must Have)
- **Description:** User identity and their listings
- **Requirements:**
 - Profile photo
 - Display name
 - My active listings
 - My saved listings

- Account settings (email, password)
- Badge: "Private Owner" vs "Dealer"
- **Acceptance Criteria:**
 - Profile loads in under 1 second
 - Users can edit their info
 - Active listings show current status

1.8 Payment Integration

- **Priority:** P0 (Must Have)
 - **Description:** Collect listing fees
 - **Requirements:**
 - Stripe integration
 - \$3 listing fee for private owners
 - Payment confirmation page
 - Listing goes live after successful payment
 - **Acceptance Criteria:**
 - Payment flow is under 30 seconds
 - Failed payments show clear error messages
 - Receipts sent via email
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Phase 2: Enhanced Features (Weeks 7-12)

2.1 Saved/Favorited Listings

- **Priority:** P1 (Should Have)
- **Description:** Users can save listings they're interested in
- **Requirements:**
 - Heart/save icon on listings
 - Saved listings section in profile
 - Notifications when saved listing is updated
- **Acceptance Criteria:**
 - Save/unsave is instant
 - No limit on saved listings
 - User can see when a saved listing is sold/removed

2.2 Notifications System

- **Priority:** P1 (Should Have)
- **Description:** Keep users engaged and informed
- **Requirements:**
 - New message notifications
 - Saved listing updates

- New listings matching interests
- Email digest option (daily/weekly)
- **Acceptance Criteria:**
 - Notifications arrive within 1 minute of trigger
 - Users can customize notification preferences
 - Unread count badge on app

2.3 Listing Analytics (for poster)

- **Priority:** P1 (Should Have)
- **Description:** Show listing performance
- **Requirements:**
 - View count
 - Save count
 - Message count
 - Simple graph over time
- **Acceptance Criteria:**
 - Analytics visible only to listing owner
 - Updates daily
 - Easy to interpret

2.4 Reporting & Moderation

- **Priority:** P1 (Should Have)
- **Description:** Safety and quality control
- **Requirements:**
 - Report listing (spam, scam, inappropriate)
 - Report user (harassment, fraud)
 - Admin dashboard to review reports
 - Ability to remove listings/ban users
- **Acceptance Criteria:**
 - Reports are reviewed within 24 hours
 - Users notified of action taken
 - Clear community guidelines

8. Feature Requirements (Future/Post-MVP)

Phase 3: Growth & Optimization (Months 4-6)

3.1 Smart Matching Algorithm

- **Description:** Suggest relevant swaps based on user preferences

- **Requirements:**
 - "Recommended for you" section
 - Based on saved listings, search history, location
 - ML model to predict good matches

3.2 Verification System

- **Description:** Build trust through verification
- **Requirements:**
 - Verified owner badge (\$10 fee)
 - Upload registration document
 - Manual review process
 - Vehicle history report integration (Carfax API)

3.3 Native Mobile Apps

- **Description:** iOS and Android apps
- **Requirements:**
 - Native performance
 - Push notifications
 - Camera integration for listing photos
 - App Store / Play Store optimization

3.4 Social Features

- **Description:** Build community
- **Requirements:**
 - Follow other users
 - Share swap success stories
 - Comment on listings
 - User reputation/reviews

3.5 Premium Subscriptions

- **Description:** Monetization tier for power users
- **Requirements:**
 - Unlimited active listings
 - Featured placement in feed
 - Advanced analytics
 - Early access to new matches
 - \$10-15/month pricing

3.6 Dealer Tier

- **Description:** Specialized features for dealers
- **Requirements:**

- Bulk listing upload
- Dealer badge
- Inventory management dashboard
- \$50-100/month pricing

3.7 Value-Added Services

- **Description:** Partner integrations
- **Requirements:**
 - Pre-purchase inspection booking
 - Title/escrow service referrals
 - Vehicle shipping quotes
 - Insurance quote integration
 - Revenue via affiliate/referral fees

3.8 Geographic Expansion

- **Description:** International markets
 - **Requirements:**
 - Multi-currency support
 - Localization (language, regulations)
 - Country-specific vehicle databases
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9. Technical Architecture

Tech Stack

Frontend:

- React + Next.js 14 (App Router)
- TypeScript
- Tailwind CSS
- Mobile-first responsive design

Backend:

- Next.js API routes (serverless)
- Supabase (PostgreSQL database, auth, storage, real-time)
- Alternative: Firebase (if real-time is critical priority)

Infrastructure:

- Vercel (hosting, CI/CD, edge functions)

- Cloudinary (image optimization, CDN)
- Stripe (payments)

Third-Party Services:

- SendGrid (transactional emails)
- Google Maps API (location, distance calculations)
- Sentry (error tracking)
- Mixpanel or PostHog (analytics)

Database Schema (Simplified)

Users Table:

- id, email, password_hash, display_name, profile_photo_url, account_type (private/dealer), created_at, location_zip

Listings Table:

- id, user_id, have_year, have_make, have_model, have_trim, have_mileage, want_description, cash_differential_min, cash_differential_max, description, location_zip, status (active/sold/expired), created_at, view_count

Photos Table:

- id, listing_id, photo_url, order, created_at

Messages Table:

- id, listing_id, sender_id, recipient_id, message_text, photo_url, created_at, read_at

Saved_Listings Table:

- id, user_id, listing_id, created_at

Security Considerations

- HTTPS everywhere
- JWT token authentication
- Rate limiting on API endpoints
- Input validation and sanitization
- SQL injection prevention (parameterized queries)
- XSS protection
- CSRF tokens
- Secure file upload (image validation, size limits)
- PII encryption at rest

- Regular dependency updates

Performance Targets

- First Contentful Paint: < 1.5s
 - Time to Interactive: < 3s
 - Lighthouse Score: 90+
 - Image load time: < 500ms per image
 - API response time: < 200ms (p95)
 - Uptime: 99.5% (MVP), 99.9% (post-launch)
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10. Go-to-Market Strategy

Pre-Launch (Weeks 1-8)

Goal: Validate demand and build initial user base

Tactics:

1. **Landing page** (Week 1)
 - "Coming Soon: Instagram for Car Swaps"
 - Email capture
 - Social proof (# of signups)
 - Share on Reddit, Facebook car groups
2. **Manual concierge test** (Weeks 2-4)
 - Find 10 people who want to swap
 - Facilitate matches manually via spreadsheet
 - Learn pain points and refine product
3. **Content creation** (Ongoing)
 - Blog: "Why Car Swaps Make Sense"
 - Instagram/TikTok: Swap stories, tips
 - Build anticipation

Success Metric: 500+ email signups before launch

Launch Phase (Weeks 9-16)

Goal: Onboard first 1,000 users and facilitate first swaps

Beta Launch (Weeks 9-12)

- Invite-only access for email list
- 50-100 users maximum
- Heavy hands-on support
- Daily check-ins and bug fixes
- Gather qualitative feedback

Public Launch (Weeks 13-16)

- Open to everyone
- Press release to automotive blogs (Jalopnik, The Drive, Autoblog)
- Reddit AMA on r/cars
- Product Hunt launch
- Paid social ads (\$500-1000 budget for testing)

Channels:

1. Organic Social:

- Reddit: r/cars, r/Autos, r/whatcarshouldIbuy, brand-specific subs
- Facebook: Car enthusiast groups (Mustang owners, JDM groups, etc.)
- Instagram: Car culture hashtags
- TikTok: Swap story content

2. Community Partnerships:

- Cars & Coffee organizers (flyers, mentions)
- Local car clubs (offer free dealer tier for club)
- YouTube car channels (potential sponsorships)

3. Content Marketing:

- SEO-optimized blog posts ("How to swap your car")
- Swap success stories
- Market data (which cars swap most)

4. Paid Acquisition:

- Facebook/Instagram ads targeting car enthusiasts
- Google Ads for "sell my [car model]" keywords
- Reddit promoted posts

Success Metrics:

- 1,000 registered users by end of Month 4
- 500 active listings

- 5 verified successful swaps
- 30% Week 1 retention

Growth Phase (Months 5-12)

Goal: Achieve product-market fit and scale to 10K users

Tactics:

1. **Referral program:** Both users get free premium month for successful referral
2. **Geographic expansion:** Launch city-by-city (start with LA, then Miami, Houston, Atlanta)
3. **Dealer outreach:** Cold email independent dealers in key markets
4. **PR push:** National automotive publications
5. **Influencer partnerships:** YouTube car reviewers, Instagram car accounts
6. **SEO investment:** Programmatic listing pages for every make/model

Success Metrics:

- 10,000 registered users
 - 100+ monthly active swaps
 - 40% Week 1 retention
 - \$10K MRR (from listings + premium subs)
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11. Risks & Mitigations

Risk Matrix

Risk	Likelihood	Impact	Mitigation
Low user adoption	Medium	High	Pre-launch validation, seed community manually
Fraud/scams	High	High	Clear disclaimers, reporting system, verification badges
Chicken-and-egg (no inventory)	High	High	Recruit dealers early, seed with own listings
Legal liability for bad swaps	Medium	High	Strong ToS, users assume all risk, insurance waiver

Competitor enters market	Low	Medium	Move fast, build community moat
Technical issues at scale	Low	Medium	Load testing, monitoring, scalable architecture
Dealer spam ruins experience	Medium	Medium	Separate feeds, rate limits, clear badges

Detailed Risk Analysis

1. Low User Adoption

Symptoms: Slow signup growth, low engagement, few listings **Root Causes:**

- Problem isn't real or addressable
- Product is too complex
- Marketing isn't reaching target audience

Mitigation:

- Pre-launch validation with landing page and manual matching
 - Start hyper-local (one city) to build density
 - Personal outreach to 50+ enthusiasts
 - Pivot features based on feedback quickly
-

2. Fraud & Scams

Symptoms: User reports of fake listings, stolen car photos, no-shows **Root Causes:**

- Low barrier to entry attracts bad actors
- Anonymous internet enables fraud
- High-value items (cars) attract scammers

Mitigation:

- Require email verification
 - Verification badge system (paid, manual review)
 - Clear "deal at your own risk" messaging
 - Report/block functionality
 - Partner with escrow service for referrals
 - Community guidelines prominently displayed
 - Consider requiring phone verification for listing
-

3. Chicken-and-Egg Problem

Symptoms: Browsers see few listings, posters get no engagement **Root Causes:**

- Network effects require critical mass
- No existing inventory at launch

Mitigation:

- Recruit 20-30 initial listers before public launch
 - Approach small dealers to seed inventory
 - Start geographic-specific (one metro area)
 - Create fake/"example" listings to show how platform works (clearly marked)
 - Cross-post to Reddit/Facebook groups to drive traffic
 - Offer free listings for first 100 users
-

4. Legal Liability

Symptoms: Users blame platform for bad swaps, potential lawsuits **Root Causes:**

- Platform facilitates transactions
- Users may assume platform guarantees quality/safety

Mitigation:

- Strong Terms of Service (lawyer reviewed)
 - Clear disclaimers on every page
 - "Platform connects users only, does not facilitate transactions"
 - Require users to acknowledge risks before posting/messaging
 - General liability insurance for the business
 - No escrow or transaction handling (keep hands-off)
 - Recommend users get pre-purchase inspections
-

5. Competitive Threats

Symptoms: Larger player launches similar product, loses market share **Root Causes:**

- No moat, easy to copy
- Incumbents have resources

Mitigation:

- Move fast and build community quickly

- Network effects as moat
 - Niche focus (enthusiasts only)
 - Superior UX/design
 - Consider white-label for larger platforms vs compete
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12. Timeline & Milestones

Phase 0: Validation (Weeks 1-2)

Owner: Founder

Activities:

- Build landing page
- Launch on Reddit/Facebook groups
- Run manual matching experiment
- Interview 20+ potential users

Milestone: 500+ email signups, 1 manual swap facilitated

Phase 1: MVP Development (Weeks 3-8)

Owner: Founder (technical)

Week 3-4:

- Set up development environment
- Database schema design
- Authentication system
- Basic UI framework

Week 5-6:

- Listing creation flow
- Browse feed
- Search & filters
- Image upload/optimization

Week 7-8:

- Messaging system

- User profiles
- Payment integration
- Testing & bug fixes

Milestone: Working MVP deployed to staging

Phase 2: Private Beta (Weeks 9-12)

Owner: Founder

Activities:

- Invite 50 users from email list
- Daily monitoring and support
- Rapid bug fixes
- Gather feedback via surveys and interviews
- Iterate on UX based on feedback

Milestone: 50 active users, 25+ listings, 3 successful swaps, 30%+ Week 1 retention

Phase 3: Public Launch (Weeks 13-16)

Owner: Founder + Marketing partner (if hired)

Activities:

- Open registration
- Press outreach
- Product Hunt launch
- Paid ads testing
- Reddit AMA
- Content marketing ramp-up

Milestone: 1,000 registered users, 500 active listings, 5 verified swaps

Phase 4: Growth (Months 5-12)

Owner: Founder + Co-founder (business)

Activities:

- Implement Phase 3 features (matching, verification, mobile apps)
- Dealer tier launch
- Geographic expansion (city by city)
- Fundraising (if needed)
- Hire first employees (customer support, marketing)

Milestone: 10,000 users, 100 monthly swaps, \$10K MRR, product-market fit

Appendix A: Open Questions

1. Should we allow test drive coordination through the platform?

- Pro: More full-service, better UX
- Con: Liability risk, complex feature
- Decision: Not for MVP, consider post-launch

2. Geographic scope: Local only or nationwide?

- Local pros: Easier swaps, stronger community
- Nationwide pros: More inventory, better matches
- Decision: Start local (one metro), expand based on demand

3. Should we moderate listings before they go live?

- Pro: Quality control, safety
- Con: Slow, resource-intensive
- Decision: Auto-publish with post-moderation and reporting

4. Pricing model for dealers?

- Subscription vs. per-listing fee
- Decision: \$50/month unlimited listings (test and iterate)

5. How to handle users who ghost after initial interest?

- Rating system? Mark as "responsive"?
 - Decision: Phase 2 feature, not MVP
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Appendix B: Competitive Analysis

Platform	Focus	Strengths	Weaknesses	Threat Level

Facebook Marketplace	General classifieds	Massive user base	Not specialized, poor UX	Medium
Craigslist	General classifieds	Free, established	Outdated, scams	Low
CarMax	Dealer trade-ins	Convenient, guaranteed	Low values	Low
Carvana	Online car buying	Slick UX, delivery	Not P2P swaps	Low
Enthusiast forums	Community	Engaged users	Fragmented, manual	Low
Bring a Trailer	Auction	High-end market	Auction model only	Low

Conclusion: No direct competitor focusing on P2P swaps with social/visual UX

Appendix C: Monetization Projection (Year 1)

Assumptions:

- Listing fee: \$3
- Premium subscription: \$15/month
- Dealer subscription: \$50/month
- Conversion rates: 10% visitor to signup, 50% signup to post, 5% upgrade to premium

Month 3 (Beta End):

- 100 users, 50 listings
- Revenue: \$150 (listings only)

Month 6:

- 1,000 users, 500 listings, 20 premium, 5 dealers
- Revenue: \$1,500 (listings) + \$300 (premium) + \$250 (dealers) = **\$2,050/month**

Month 12:

- 10,000 users, 3,000 listings, 200 premium, 30 dealers
- Revenue: \$9,000 (listings) + \$3,000 (premium) + \$1,500 (dealers) = **\$13,500/month**

Year 1 Total: ~\$60,000 revenue (ramp-up curve)

Appendix D: Success Criteria for Continuation

After Beta (Month 3):

-  Continue if: 30%+ Week 1 retention, 3+ successful swaps, NPS > 40
-  Kill if: <20% retention, no swaps, users complain product isn't useful

After Public Launch (Month 6):

-  Continue if: 1,000+ users, 10+ monthly swaps, \$2K+ MRR, clear PMF signals
-  Pivot if: Slow growth, low engagement, can't monetize

Year End (Month 12):

-  Scale if: 10K+ users, 100+ monthly swaps, \$10K+ MRR, viral growth coefficient >1
-  Shutdown if: Can't achieve sustainable growth or monetization

Document History

Version	Date	Author	Changes
1.0	Jan 10, 2026	[Founder]	Initial draft

Next Steps:

1. Review and approve PRD
2. Begin validation phase (landing page)
3. Set up development environment
4. Start MVP build

Questions or Feedback: [Your Email]

Tech Architecture

Tech Stack

Frontend

- React 18 with Vite (faster than Next.js, simpler for your use case)
- TypeScript
- Tailwind CSS
- React Router (client-side routing)
- Axios or Fetch API (HTTP calls)
- Deployed to: AWS S3 + CloudFront

Backend

- Java 17+ with Spring Boot 3.2
- Spring Web (REST APIs)
- Spring Security (authentication)
- Spring Data JPA (ORM)
- JWT (JSON Web Tokens) via **j jwt** library
- Hibernate (ORM implementation)
- Maven or Gradle (build tool)
- Docker (containerization)
- Deployed to: AWS ECS Fargate

Database

- AWS RDS PostgreSQL 15
 - Instance: **db.t3.micro** (1 vCPU, 1GB RAM) → ~\$15/month
 - Storage: 20GB SSD
 - Automated backups (7-day retention)
 - Multi-AZ for production (high availability)

Storage

- AWS S3 (image storage)
 - Bucket for user-uploaded car photos
 - Lifecycle policy: delete images after listing expires (90 days)
- AWS CloudFront (CDN)
 - Global content delivery

- Cache images for fast loading
- HTTPS by default

Caching (*Optional for MVP - can add later*)

- **AWS ElastiCache Redis**
 - Cache hot listings
 - Session storage
 - Rate limiting
 - Instance: **cache.t3.micro** → ~\$15/month

Email

- **AWS SES (Simple Email Service)**
 - Transactional emails (verification, password reset, notifications)
 - Cost: \$0.10 per 1,000 emails
 - Alternative: SendGrid (easier setup, 100 emails/day free)

Payments

- **Stripe**
 - \$3-5 listing fees
 - Subscription billing (premium/dealer tiers)
 - PCI compliant

Monitoring & Logging

- **AWS CloudWatch**
 - Application logs (ECS logs)
 - Metrics (CPU, memory, API latency)
 - Alarms (error rate threshold alerts)
- **AWS CloudWatch Logs Insights (query logs)**
- **Sentry (*Optional*)** - Error tracking, free tier covers MVP

CI/CD

- **GitHub Actions**
 - Build Spring Boot JAR
 - Build Docker image

- **Push to AWS ECR (Elastic Container Registry)**
- **Deploy to ECS Fargate**
- **Frontend: Build React → Upload to S3 → Invalidate CloudFront cache**

Infrastructure as Code (*Optional but recommended*)

- **Terraform or AWS CloudFormation**
 - **Version control your AWS infrastructure**
 - **Repeatable deployments**
 - **Easy to spin up staging/production environments**

Third-Party Services

- **Google Maps API (location, distance calculations)**
- **Stripe API (payments)**
- **AWS SDK for Java (S3, SES integration)**

Tech Stack

Frontend:

- React + Next.js 14 (App Router)
- TypeScript
- Tailwind CSS
- Mobile-first responsive design

Backend:

- Next.js API routes (serverless)
- Supabase (PostgreSQL database, auth, storage, real-time)
- Alternative: Firebase (if real-time is critical priority)

Infrastructure:

- Vercel (hosting, CI/CD, edge functions)
- Cloudinary (image optimization, CDN)

- Stripe (payments)

Third-Party Services:

- SendGrid (transactional emails)
- Google Maps API (location, distance calculations)
- Sentry (error tracking)
- Mixpanel or PostHog (analytics)

Database Schema (Simplified)

Users Table:

- id, email, password_hash, display_name, profile_photo_url, account_type (private/dealer), created_at, location_zip

Listings Table:

- id, user_id, have_year, have_make, have_model, have_trim, have_mileage, want_description, cash_differential_min, cash_differential_max, description, location_zip, status (active/sold/expired), created_at, view_count

Photos Table:

- id, listing_id, photo_url, order, created_at

Messages Table:

- id, listing_id, sender_id, recipient_id, message_text, photo_url, created_at, read_at

Saved_Listings Table:

- id, user_id, listing_id, created_at

Security Considerations

- HTTPS everywhere
- JWT token authentication
- Rate limiting on API endpoints
- Input validation and sanitization
- SQL injection prevention (parameterized queries)
- XSS protection
- CSRF tokens
- Secure file upload (image validation, size limits)
- PII encryption at rest
- Regular dependency updates

Performance Targets

- First Contentful Paint: < 1.5s
- Time to Interactive: < 3s
- Lighthouse Score: 90+
- Image load time: < 500ms per image
- API response time: < 200ms (p95)
- Uptime: 99.5% (MVP), 99.9% (post-launch)

Tech Stack

Frontend

- **React 18** with **Vite** (faster than Next.js, simpler for your use case)
- **TypeScript**
- **Tailwind CSS**
- **React Router** (client-side routing)
- **Axios** or **Fetch API** (HTTP calls)
- **Deployed to:** AWS S3 + CloudFront

Backend

- **Java 17+** with **Spring Boot 3.2**
- **Spring Web** (REST APIs)
- **Spring Security** (authentication)
- **Spring Data JPA** (ORM)
- **JWT** (JSON Web Tokens) via **jjwt** library
- **Hibernate** (ORM implementation)
- **Maven** or **Gradle** (build tool)
- **Docker** (containerization)
- **Deployed to:** AWS ECS Fargate + API Gateway

Database

- **AWS RDS PostgreSQL 15**
 - Instance: **db.t3.micro** (1 vCPU, 1GB RAM) → ~\$15/month
 - Storage: 20GB SSD
 - Automated backups (7-day retention)
 - Multi-AZ for production (high availability)

Storage

- **AWS S3** (image storage)
 - Bucket for user-uploaded car photos
 - Lifecycle policy: delete images after listing expires (90 days)
- **AWS CloudFront** (CDN)
 - Global content delivery
 - Cache images for fast loading
 - HTTPS by default

Caching (*Optional for MVP - can add later*)

- **AWS ElastiCache Redis**
 - Cache hot listings

- Session storage
- Rate limiting
- Instance: `cache.t3.micro` → ~\$15/month

Email

- **AWS SES** (Simple Email Service)
 - Transactional emails (verification, password reset, notifications)
 - Cost: \$0.10 per 1,000 emails
 - **Alternative:** SendGrid (easier setup, 100 emails/day free)

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Infrastructure as Code (*Optional but recommended*)

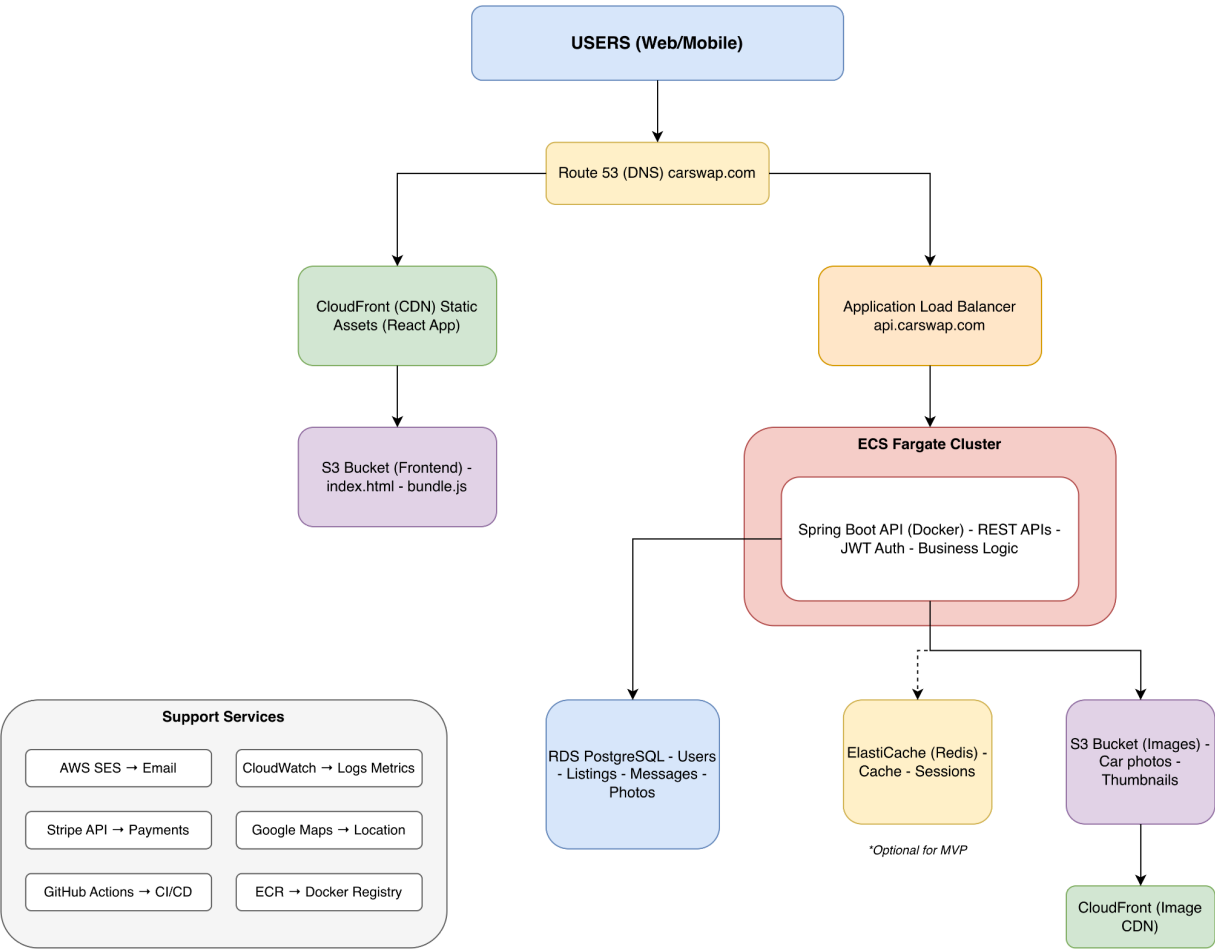
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- **Stripe API** (payments)
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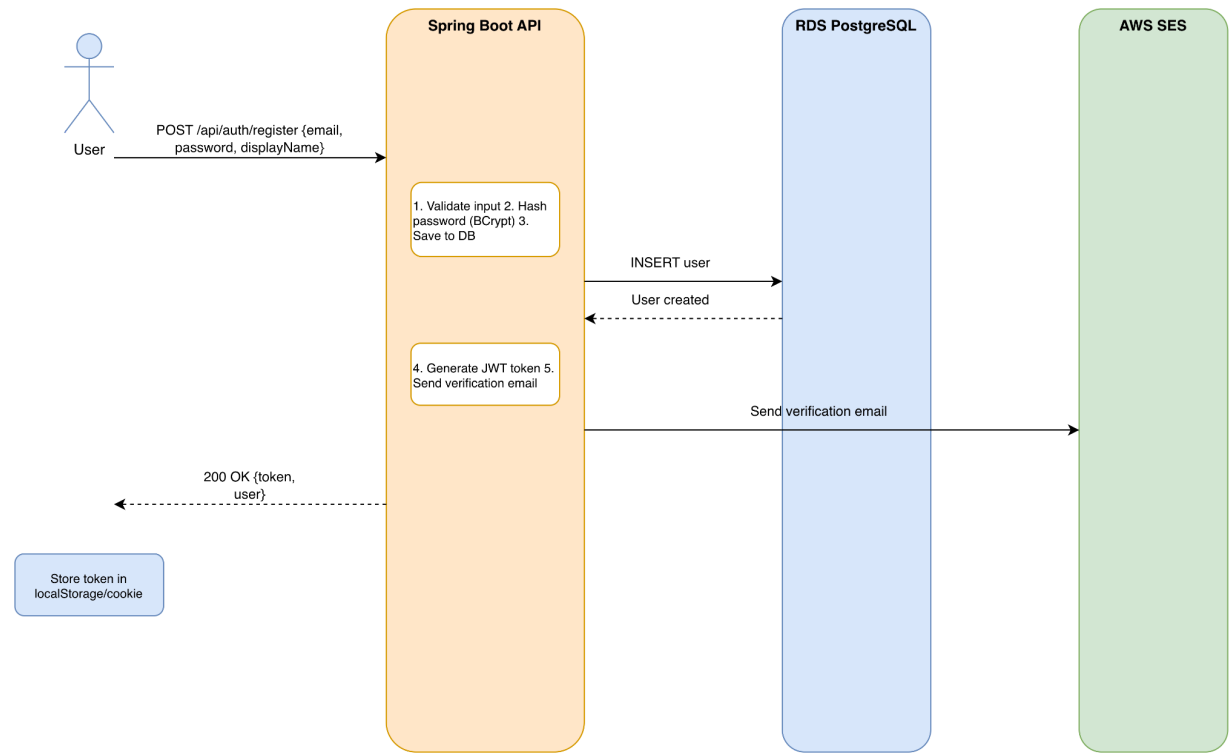
System Architecture Diagram

System Architecture Diagram



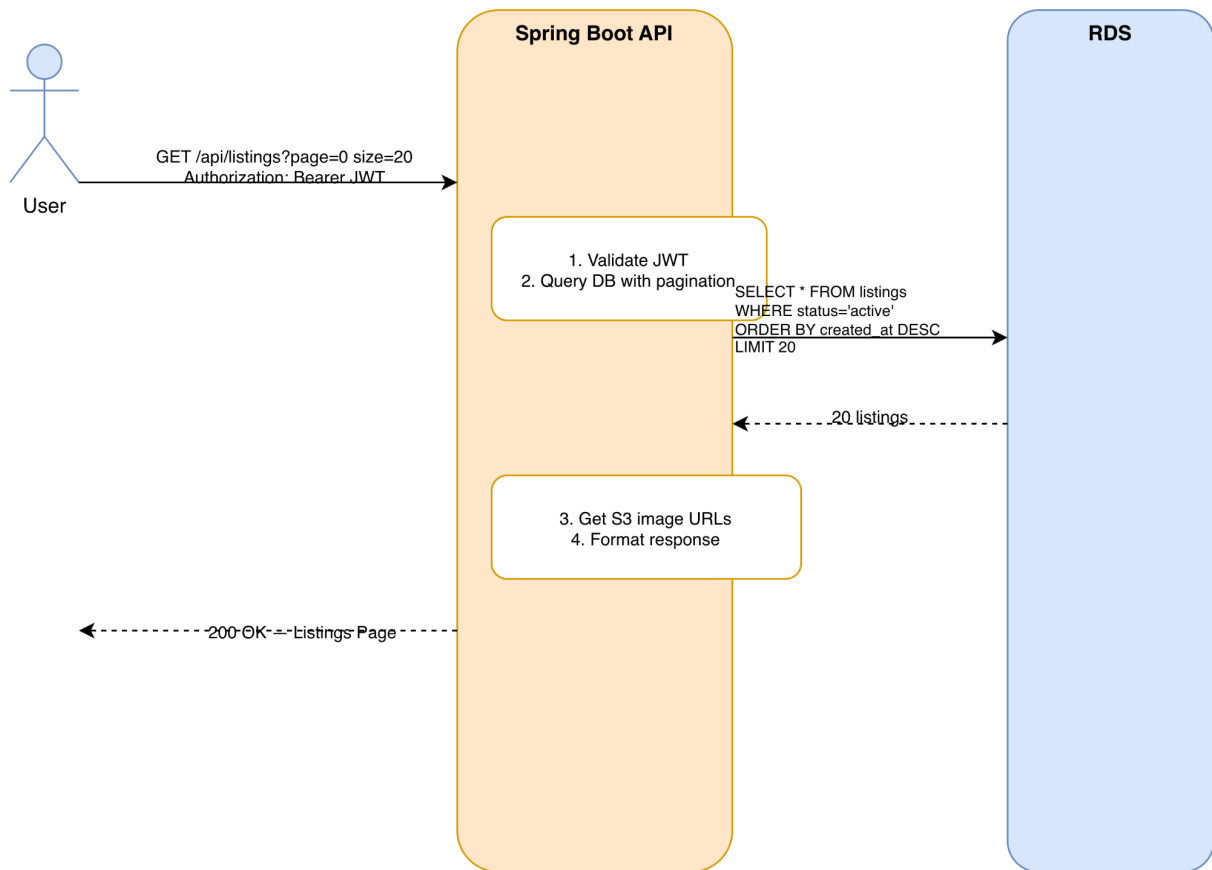
User Registration Flow

User Registration Flow



Browse Listings Flow

Browse Listings Flow



Messaging Flow

Messaging Flow

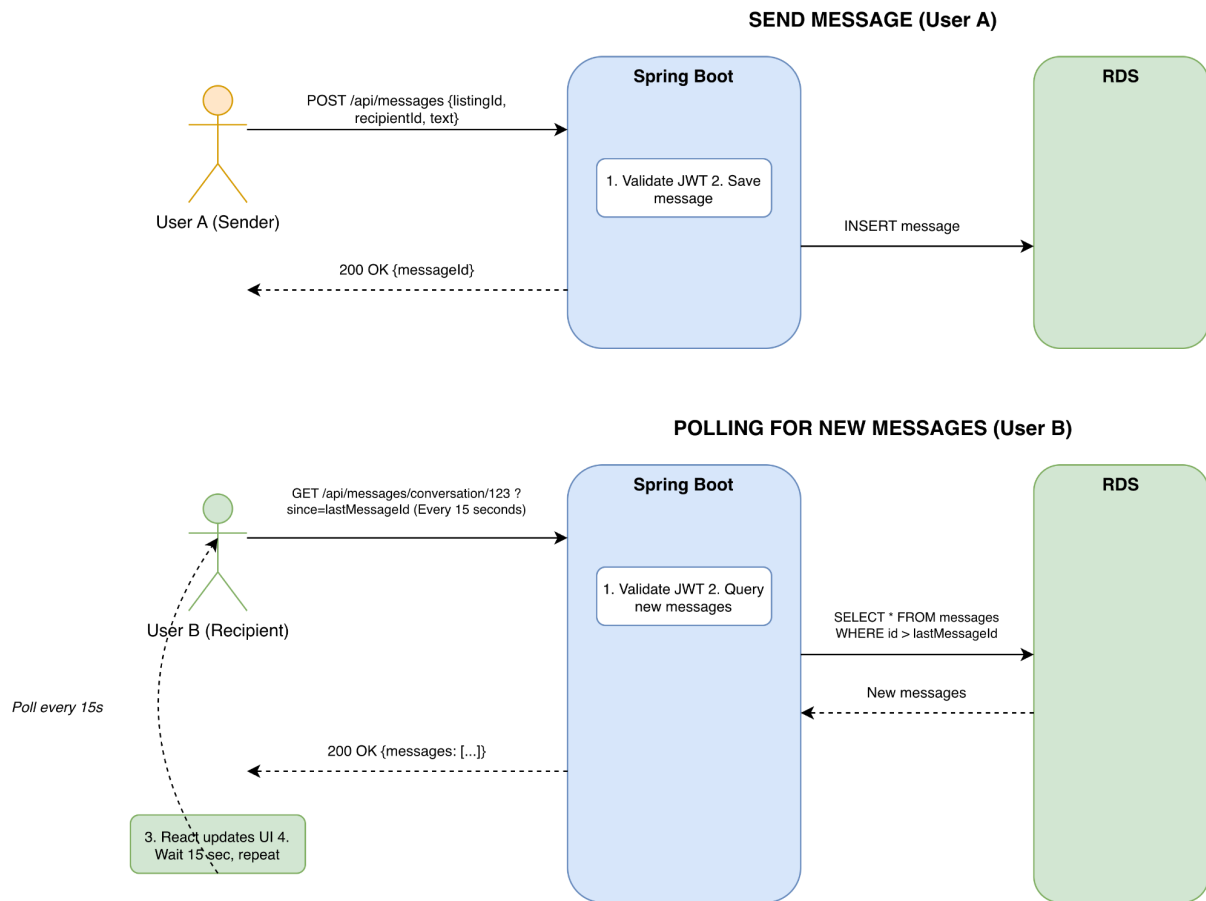
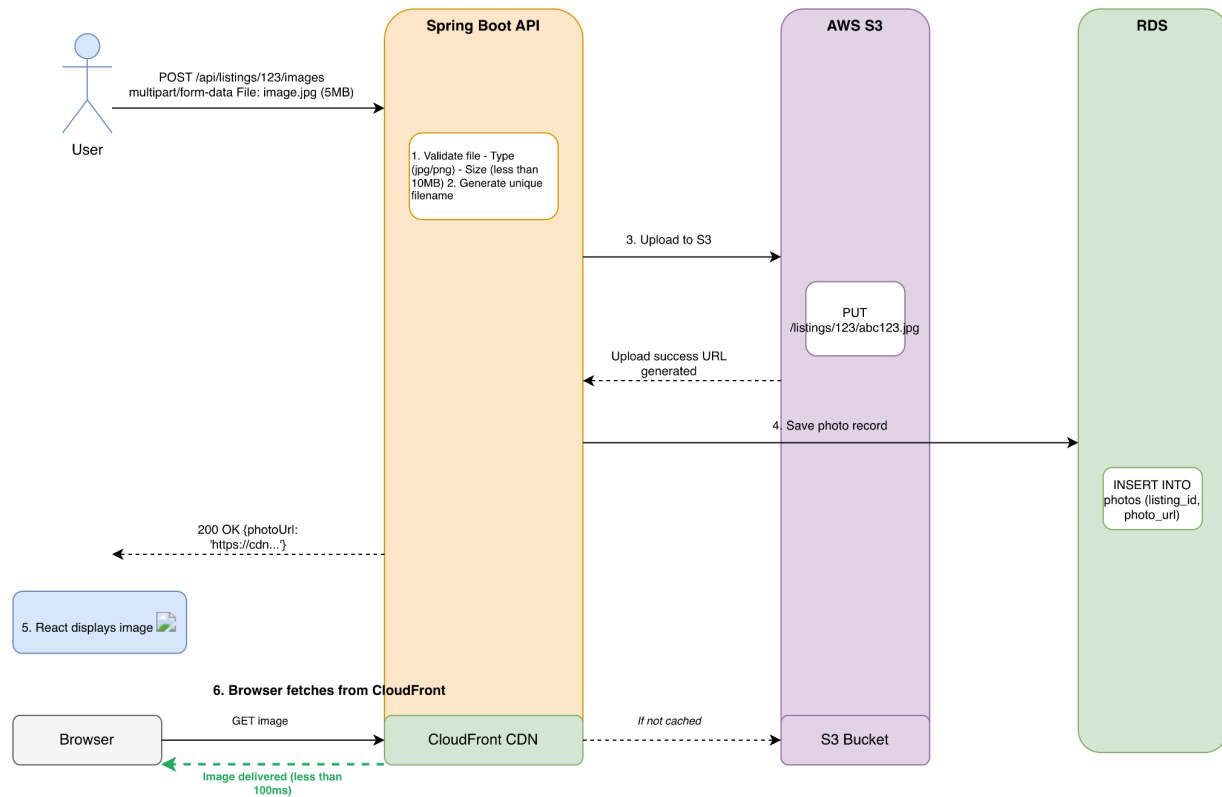


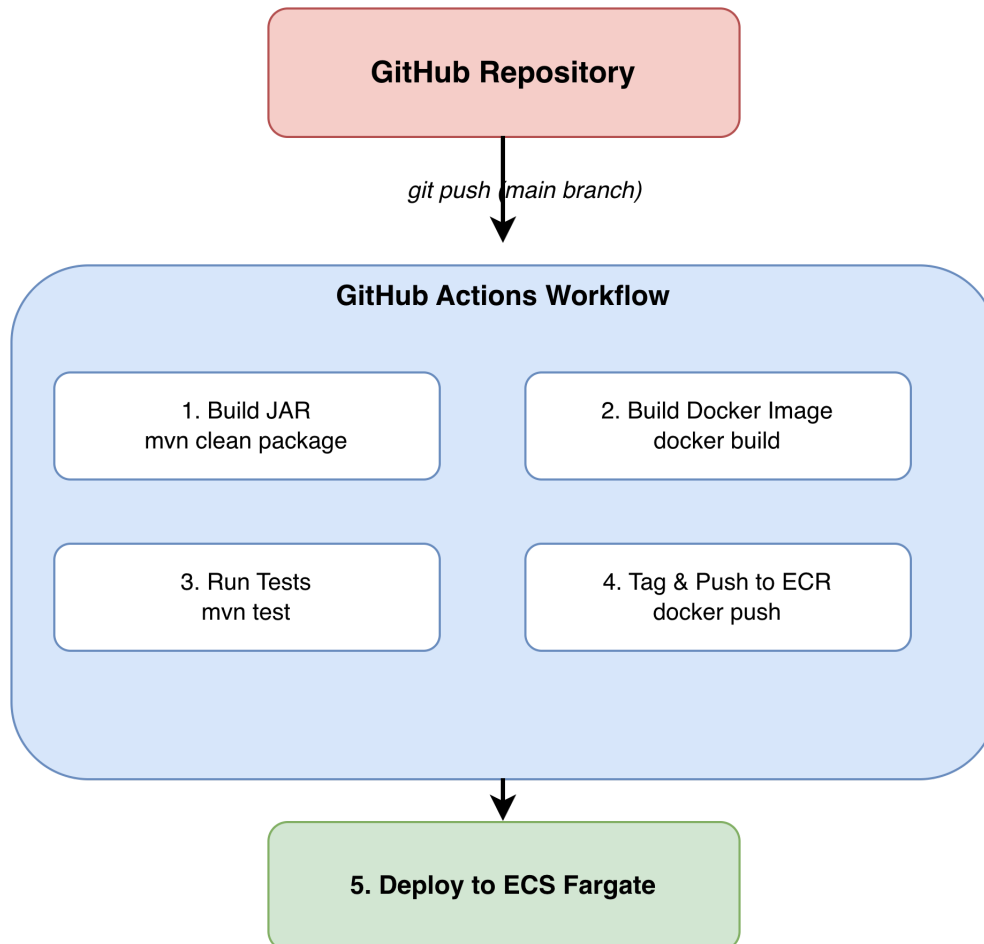
Image Upload & Delivery Flow

5. Image Upload & Delivery Flow



CI/CD Deployment Pipeline

CI/CD Deployment Pipeline



MVP - Phase 1

Phase 1: Core Platform (Weeks 1-6)

1.1 User Authentication

- **Priority:** P0 (Must Have)
- **Description:** Users can create accounts and log in
- **Requirements:**
 - Email/password registration
 - Google OAuth sign-in
 - Email verification
 - Password reset flow
- **Acceptance Criteria:**
 - User can create account in under 60 seconds
 - Session persists across browser sessions
 - Secure authentication (hashed passwords, JWT tokens)

1.2 Create Swap Listing

- **Priority:** P0 (Must Have)
- **Description:** Users can post what they have and what they want
- **Requirements:**
 - Upload 3-10 photos (min 3 required)
 - Enter vehicle details:
 - What I have: Year, Make, Model, Trim, Mileage, Location (city/state)
 - What I want: Make, Model preferences (open text)
 - Cash differential range (e.g., willing to add \$5K-\$10K)
 - Description field (500 char max for MVP)
 - Location-based (zip code for general area)
- **Acceptance Criteria:**
 - Listing creation takes under 3 minutes
 - Photos are compressed/optimized automatically
 - Location is validated against US zip codes
 - Listing goes live immediately after payment

1.3 Browse Feed

- **Priority:** P0 (Must Have)
- **Description:** Instagram-style visual feed of swap listings
- **Requirements:**
 - Infinite scroll of listings
 - Large, prominent photos
 - Key info overlay: Make/Model, Location, "Want: [X]"
 - Default sort: Most recent first
 - Works on mobile and desktop

- **Acceptance Criteria:**
 - Feed loads in under 2 seconds
 - Images are high quality but optimized
 - Smooth scrolling on mobile
 - No listings older than 90 days shown

1.4 Basic Search & Filters

- **Priority:** P0 (Must Have)
- **Description:** Users can find relevant listings
- **Requirements:**
 - Search by make/model
 - Filter by:
 - Location (within X miles of zip code)
 - Vehicle type (sedan, SUV, sports car, truck)
 - Price range (cash differential)
 - Clear all filters option
- **Acceptance Criteria:**
 - Search returns results in under 1 second
 - Filters can be combined
 - Results update without page refresh

1.5 Listing Detail View

- **Priority:** P0 (Must Have)
- **Description:** Full information about a specific listing
- **Requirements:**
 - Photo gallery (swipeable on mobile)
 - Complete vehicle details
 - Seller profile preview
 - "Message Seller" button
 - "Save" button
 - Posted date
 - View count
- **Acceptance Criteria:**
 - All images load progressively
 - Contact button is always visible
 - Works seamlessly on mobile

1.6 Direct Messaging

- **Priority:** P0 (Must Have)
- **Description:** Users can communicate about potential swaps
- **Requirements:**
 - Real-time chat interface

- Message threads per listing
- Push notifications for new messages (web/mobile)
- Photo sharing in messages
- Report/block functionality
- **Acceptance Criteria:**
 - Messages deliver within 2 seconds
 - Users can see when other person is typing
 - Message history is preserved
 - Blocked users cannot contact you

1.7 User Profile

- **Priority:** P0 (Must Have)
- **Description:** User identity and their listings
- **Requirements:**
 - Profile photo
 - Display name
 - My active listings
 - My saved listings
 - Account settings (email, password)
 - Badge: "Private Owner" vs "Dealer"
- **Acceptance Criteria:**
 - Profile loads in under 1 second
 - Users can edit their info
 - Active listings show current status

1.8 Payment Integration

- **Priority:** P0 (Must Have)
- **Description:** Collect listing fees
- **Requirements:**
 - Stripe integration
 - \$3 listing fee for private owners
 - Payment confirmation page
 - Listing goes live after successful payment
- **Acceptance Criteria:**
 - Payment flow is under 30 seconds
 - Failed payments show clear error messages
 - Receipts sent via email

MVP - Phase 2

Phase 2: Enhanced Features (Weeks 7-12)

2.1 Saved/Favorited Listings

- **Priority:** P1 (Should Have)
- **Description:** Users can save listings they're interested in
- **Requirements:**
 - Heart/save icon on listings
 - Saved listings section in profile
 - Notifications when saved listing is updated
- **Acceptance Criteria:**
 - Save/unsave is instant
 - No limit on saved listings
 - User can see when a saved listing is sold/removed

2.2 Notifications System

- **Priority:** P1 (Should Have)
- **Description:** Keep users engaged and informed
- **Requirements:**
 - New message notifications
 - Saved listing updates
 - New listings matching interests
 - Email digest option (daily/weekly)
- **Acceptance Criteria:**
 - Notifications arrive within 1 minute of trigger
 - Users can customize notification preferences
 - Unread count badge on app

2.3 Listing Analytics (for poster)

- **Priority:** P1 (Should Have)
- **Description:** Show listing performance
- **Requirements:**
 - View count
 - Save count
 - Message count
 - Simple graph over time
- **Acceptance Criteria:**
 - Analytics visible only to listing owner
 - Updates daily
 - Easy to interpret

2.4 Reporting & Moderation

- **Priority:** P1 (Should Have)
- **Description:** Safety and quality control
- **Requirements:**
 - Report listing (spam, scam, inappropriate)
 - Report user (harassment, fraud)
 - Admin dashboard to review reports
 - Ability to remove listings/ban users
- **Acceptance Criteria:**
 - Reports are reviewed within 24 hours
 - Users notified of action taken
 - Clear community guidelines

Post-MVP

Phase 3: Growth & Optimization (Months 4-6)

3.1 Smart Matching Algorithm

- **Description:** Suggest relevant swaps based on user preferences
- **Requirements:**
 - "Recommended for you" section
 - Based on saved listings, search history, location
 - ML model to predict good matches

3.2 Verification System

- **Description:** Build trust through verification
- **Requirements:**
 - Verified owner badge (\$10 fee)
 - Upload registration document
 - Manual review process
 - Vehicle history report integration (Carfax API)

3.3 Native Mobile Apps

- **Description:** iOS and Android apps
- **Requirements:**
 - Native performance
 - Push notifications
 - Camera integration for listing photos
 - App Store / Play Store optimization

3.4 Social Features

- **Description:** Build community
- **Requirements:**
 - Follow other users
 - Share swap success stories
 - Comment on listings
 - User reputation/reviews

3.5 Premium Subscriptions

- **Description:** Monetization tier for power users
- **Requirements:**
 - Unlimited active listings
 - Featured placement in feed
 - Advanced analytics
 - Early access to new matches

- \$10-15/month pricing

3.6 Dealer Tier

- **Description:** Specialized features for dealers
- **Requirements:**
 - Bulk listing upload
 - Dealer badge
 - Inventory management dashboard
 - \$50-100/month pricing

3.7 Value-Added Services

- **Description:** Partner integrations
- **Requirements:**
 - Pre-purchase inspection booking
 - Title/escrow service referrals
 - Vehicle shipping quotes
 - Insurance quote integration
 - Revenue via affiliate/referral fees

3.8 Geographic Expansion

- **Description:** International markets
- **Requirements:**
 - Multi-currency support
 - Localization (language, regulations)
 - Country-specific vehicle databases

Go to Market Strategy

Pre-Launch (Weeks 1-8)

Goal: Validate demand and build initial user base

Tactics:

1. **Landing page** (Week 1)
 - "Coming Soon: Instagram for Car Swaps"
 - Email capture
 - Social proof (# of signups)
 - Share on Reddit, Facebook car groups
2. **Manual concierge test** (Weeks 2-4)
 - Find 10 people who want to swap
 - Facilitate matches manually via spreadsheet
 - Learn pain points and refine product
3. **Content creation** (Ongoing)
 - Blog: "Why Car Swaps Make Sense"
 - Instagram/TikTok: Swap stories, tips
 - Build anticipation

Success Metric: 500+ email signups before launch

Launch Phase (Weeks 9-16)

Goal: Onboard first 1,000 users and facilitate first swaps

Beta Launch (Weeks 9-12)

- Invite-only access for email list
- 50-100 users maximum
- Heavy hands-on support
- Daily check-ins and bug fixes
- Gather qualitative feedback

Public Launch (Weeks 13-16)

- Open to everyone
- Press release to automotive blogs (Jalopnik, The Drive, Autoblog)
- Reddit AMA on r/cars
- Product Hunt launch

- Paid social ads (\$500-1000 budget for testing)

Channels:

1. Organic Social:

- Reddit: r/cars, r/Autos, r/whatcarshouldlbuy, brand-specific subs
- Facebook: Car enthusiast groups (Mustang owners, JDM groups, etc.)
- Instagram: Car culture hashtags
- TikTok: Swap story content

2. Community Partnerships:

- Cars & Coffee organizers (flyers, mentions)
- Local car clubs (offer free dealer tier for club)
- YouTube car channels (potential sponsorships)

3. Content Marketing:

- SEO-optimized blog posts ("How to swap your car")
- Swap success stories
- Market data (which cars swap most)

4. Paid Acquisition:

- Facebook/Instagram ads targeting car enthusiasts
- Google Ads for "sell my [car model]" keywords
- Reddit promoted posts

Success Metrics:

- 1,000 registered users by end of Month 4
- 500 active listings
- 5 verified successful swaps
- 30% Week 1 retention

Growth Phase (Months 5-12)

Goal: Achieve product-market fit and scale to 10K users

Tactics:

1. **Referral program:** Both users get free premium month for successful referral
2. **Geographic expansion:** Launch city-by-city (start with LA, then Miami, Houston, Atlanta)
3. **Dealer outreach:** Cold email independent dealers in key markets

4. **PR push:** National automotive publications
5. **Influencer partnerships:** YouTube car reviewers, Instagram car accounts
6. **SEO investment:** Programmatic listing pages for every make/model

Success Metrics:

- 10,000 registered users
- 100+ monthly active swaps
- 40% Week 1 retention
- \$10K MRR (from listings + premium subs)